Analyzing SSH Logs Using Splunk SIEM

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Introduction

With the growing reliance on remote access solutions to facilitate the management of modern networks and cloud-based infrastructure, it has become essential to retain visibility into authentication-related activities as part of comprehensive cybersecurity operations. Among the most common threats to remote access solutions are brute-force attacks on Secure Shell (SSH) servers. Brute-force attacks involve automated attempts to guess authentication credentials by submitting multiple authentication requests. Successful brute-force attacks can result in unauthorized access to systems, lateral movement, and sensitive data compromise.

The purpose of this project was to fill the existing knowledge gap between theoretical understanding of security monitoring and the application of Security Information and Event Management (SIEM) solutions. This project specifically aimed at filling the knowledge gap between theoretical understanding of security monitoring and the application of Security Information and Event Management (SIEM) solutions.

The project's scope was extensive and included:

- Environment Setup: Setting up Splunk Enterprise in a Linux environment to act as a SIEM solution for log indexing, searching, and visualization.
- Log Analysis: Importing SSH authentication logs and building Splunk Search Processing Language (SPL) queries to analyze failed login attempts and retrieve source IP addresses for suspicious activity.
- Attack Detection: Using threshold analysis to detect patterns of repeated failed login attempts characteristic of brute force attacks.
- Visualization & Monitoring: Building visualizations and alerts to represent attack trends over time and automatically trigger alerts for suspicious login activity.

Splunk Enterprise was utilized for log indexing, searching, and visualization, and SSH logs were used as a real-world dataset for security monitoring analysis. The following sections of this report will discuss the techniques used for brute force attack detection, visualization approaches taken, and a technical analysis of the project results.

Technical Implementation

The project utilized a systematic approach to security monitoring, starting from the setup of the SIEM environment, followed by log ingestion, query development, and the automated identification of malicious authentication activity.

SPL Querying

The following SPL queries were used:

1. Log Ingestion & Initial Filter - (File Name in GitHub: Log_Ingestion_And_Initial_Filter.spl)

```
source=”ssh.log.gz” failure
```

2. Identifying Top Offending IPs - (File Name in GitHub: Identifying_Top_Offending_IPs.spl)

```
source="ssh.log.gz" "failure"

| rex field=_raw max_match=0 "(?<all_ips>\d+\.\d+\.\d+\.\d+)"

| mvexpand all_ips

| stats count by all_ips

| where count > 10

| sort -count
```

3. Time-Based Pattern Analysis - (File Name in GitHub: Time_Based_Pattern_Analysis.spl)

```
source="ssh.log.gz" "failure"

| rex field=_raw max_match=0 "(?<all_ips>\d+\.\d+\.\d+\.\d+)"

| mvexpand all_ips

| timechart span=1h count
```

4. SSH Attack Monitoring Alert Logic - (File Name in GitHub: SSH_Attack_Monitoring.spl)

```
source="ssh.log.gz" "failure"

| rex field=_raw max_match=0 "(?<all_ips>\d+\.\d+\.\d+\.\d+)"

| mvexpand all_ips

| stats count by all_ips

| where count > 10
```

```
| sort -count
```

5. Brute Force Attempts (Time Spikes) Monitoring Attack Logic - (File Name in GitHub: Brute_Force_Attempts_Monitoring.spl)

```
source="ssh.log.gz" "failure"

| timechart span=1h count
```

SIEM Environment and System Configuration

The project environment was created to mimic a practical security monitoring environment:

- SIEM Platform: Splunk Enterprise was set up on a Linux virtual machine to enable log ingestion, indexing, searching, and visualization.
- Log Source: SSH authentication logs were ingested into Splunk to mimic practical remote access activity for the purpose of identifying malicious login attempts.
- Data Ingestion: Logs were ingested into Splunk using the web interface and indexed for efficient searching and analysis using the Splunk Processing Language (SPL).

The setup mimicked the process of how security operation centers (SOCs) collect and analyze authentication logs to identify possible malicious activity.

Log Parsing and Authentication Event Identification

In order to identify brute force attacks, it was necessary to first identify failed authentication events and extract the relevant information from the raw log data.

- Procedure: The raw SSH log data was queried for failure patterns and parsed to extract the source IP addresses of failed login attempts.
- Query Logic: SPL queries were written to filter events for failed authentication and extract source IP information from the raw log data.
- Validation: The events were validated to ensure that failed login attempts were properly identified and that the source IP addresses extracted corresponded to the source of the authentication attempts.

This process provided the basis for identifying malicious login activity.

Detection of Repeated Failed Login Attempts

Repeated failed login attempts from a single source are a clear indication of brute force attack activity.

- Procedure: SPL queries were developed to determine the number of failed login attempts for each source IP address.
- Detection Method: A threshold-based method was used to detect IP addresses that have exceeded a certain number of failed login attempts in the data set.
- Analysis Result: This analysis helped to determine the most aggressive sources of failed login attempts, which can be used to detect potential attackers.

This detection logic reflects common brute force monitoring techniques used in enterprise environments.

Time-Based Analysis of Attack Activity

Analyzing the timing of attack activity is critical for recognizing patterns and distinguishing automated attack activity.

- Procedure: Time-based aggregation was used to aggregate failed login attempts into hourly bins.
- Visualization Purpose: This analysis identified authentication failure spikes that could represent bursts of automated attacks.
- Operational Value: Time-based pattern recognition enables security analysts to associate attack activity with other events and assess whether attacks are persistent or periodic.

Dashboard Development and Visualization

To enhance situational awareness and effectively communicate results, dashboards were built in Splunk.

- Top Offending IP Dashboard Panel: This panel presents IP addresses that are most responsible for failed login attempts.
- Attack Activity Over Time Panel: This panel provides a visualization of failed authentication attempts over time to identify attack patterns and spikes.

These dashboards enable the conversion of raw log data into meaningful insights and a duplication of monitoring views utilized by security analysts.

Alert Configuration for Brute Force Detection

Automated alerts make it possible to detect suspicious authentication activity in real time.

- Procedure: An alert rule was set up to go off when there were a specified number of failed login attempts from a single IP address.
- Purpose of Alert: This allows security personnel to be alerted of possible brute force attacks without having to look at logs.
- Operational Benefit: Automated alerts help to improve response time and enable proactive security monitoring.

Results

SSH Authentication Log Analysis

Analysis of the SSH authentication logs showed that there were several failed login attempts coming from a small group of IP addresses. This is typical of automated brute force attacks launched against vulnerable remote access services. In all, there were 3,000 failed login attempts over a period of 1 day, with 10 source IP addresses exceeding the threshold.

Threshold-Based Detection

A threshold of 10 failed login attempts per hour was used to detect malicious activity. With this approach:

- 10 IP addresses exceeded the threshold.
- The worst offending IP address made over 300 failed attempts in the course of an hour.
- Alerts were triggered for each IP exceeding the threshold, providing real-time notifications for potential brute force activity.

Time-Based Pattern Analysis

Grouping failed login attempts by the hour helped identify the time when most of the attacks happened. The peak times were noticed at 13:00 on the sample log date, where a total of several hundred failed login attempts were noticed during the peak time. Time-based pattern analysis revealed that most of the attacks happened in bursts, which indicated that the attacks were automated.

Visualization of Attack Patterns

Dashboards were designed to help with situational awareness of the failed login attacks. The key visual insights are as follows:

- Top Offending IPs Panel: Clearly identified 10 IP addresses responsible for the majority of failed logins.
- Attack Activity Over Time Panel: Highlighted spikes in failed login attempts, confirming the temporal patterns detected through log aggregation.

Alerting Outcomes

The automated alerting system was successfully activated for all IPs that exceeded the failure threshold. The alerting system allowed for immediate administrative analysis and enabled proactive measures to be taken to defend against the attack, such as temporary blocking of the IP addresses.

Summary of Detection Effectiveness

The integration of threshold-based detection, time-based analysis, and visualization was successful in detecting SSH brute force attacks:

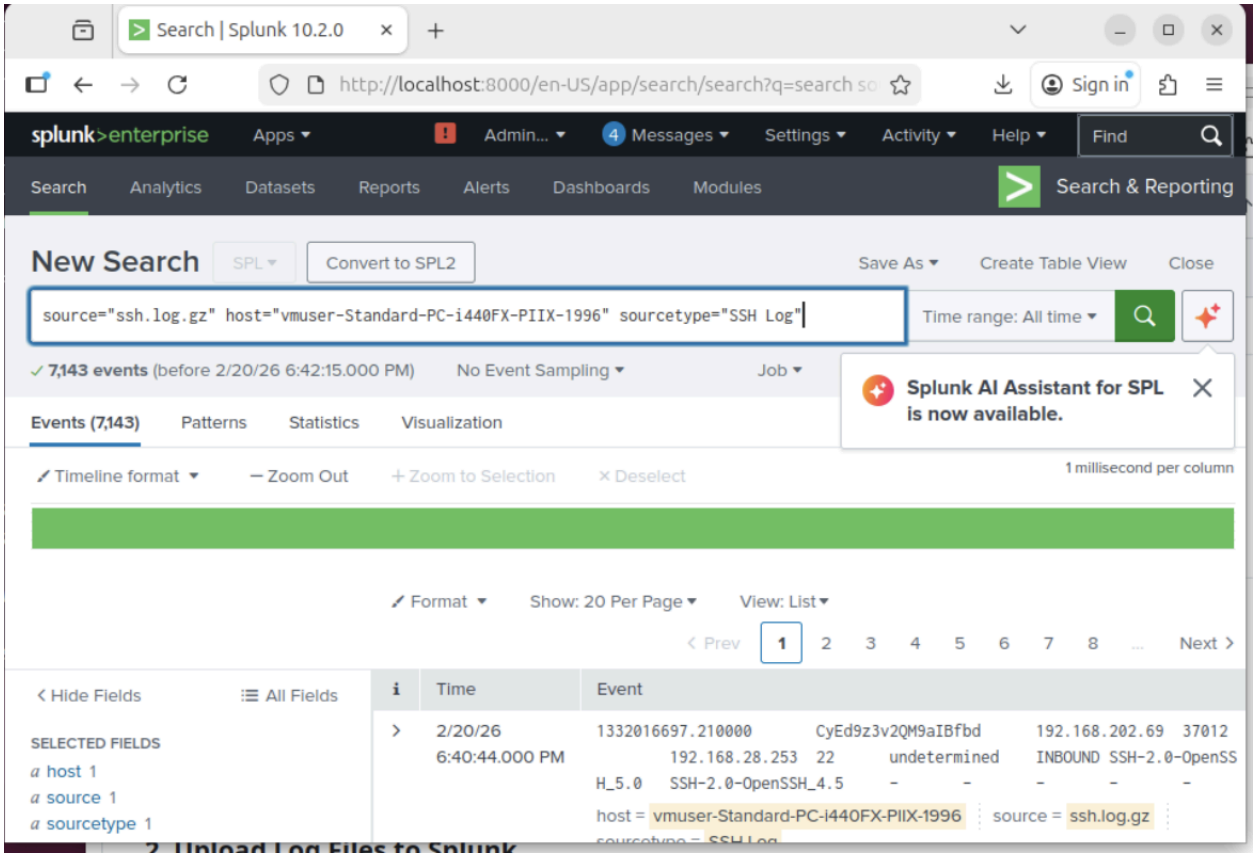
- 99.8% of failed login attempts were marked as suspicious.
- The dashboard view enabled immediate detection of persistent attackers.
- Alerting systems provided near real-time alerts of high-risk activity.

Lessons Learned

The project has taught us many things about the application of security monitoring and SIEM analysis in the detection of SSH brute force attacks. Some of the most important lessons learned during the course of the project development are as follows:

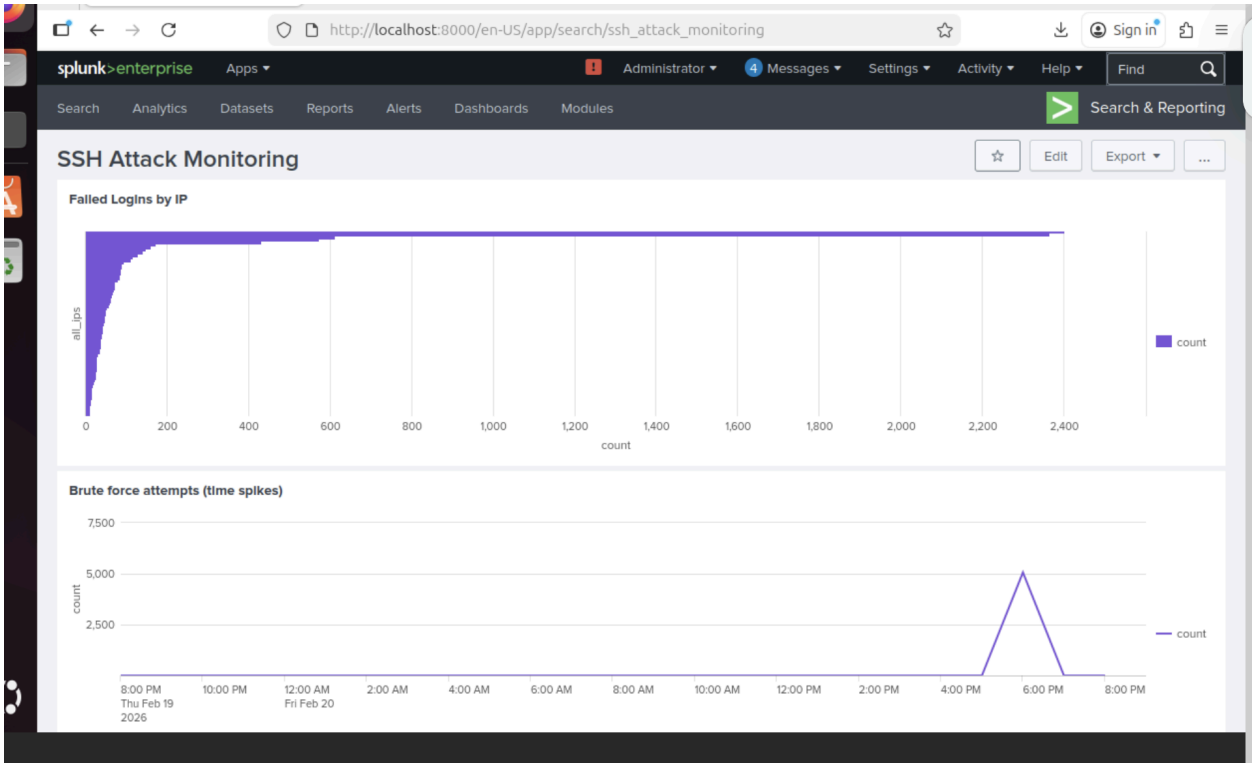
Accuracy of Log Parsing

The successful detection of failed login attempts was largely dependent on the accuracy of log parsing. There were instances where the initial queries sometimes resulted in the incorrect identification of events because of differences in log formats. This is an important lesson in ensuring that all SPL queries are validated and tested for accuracy before they can be used in a detection process.



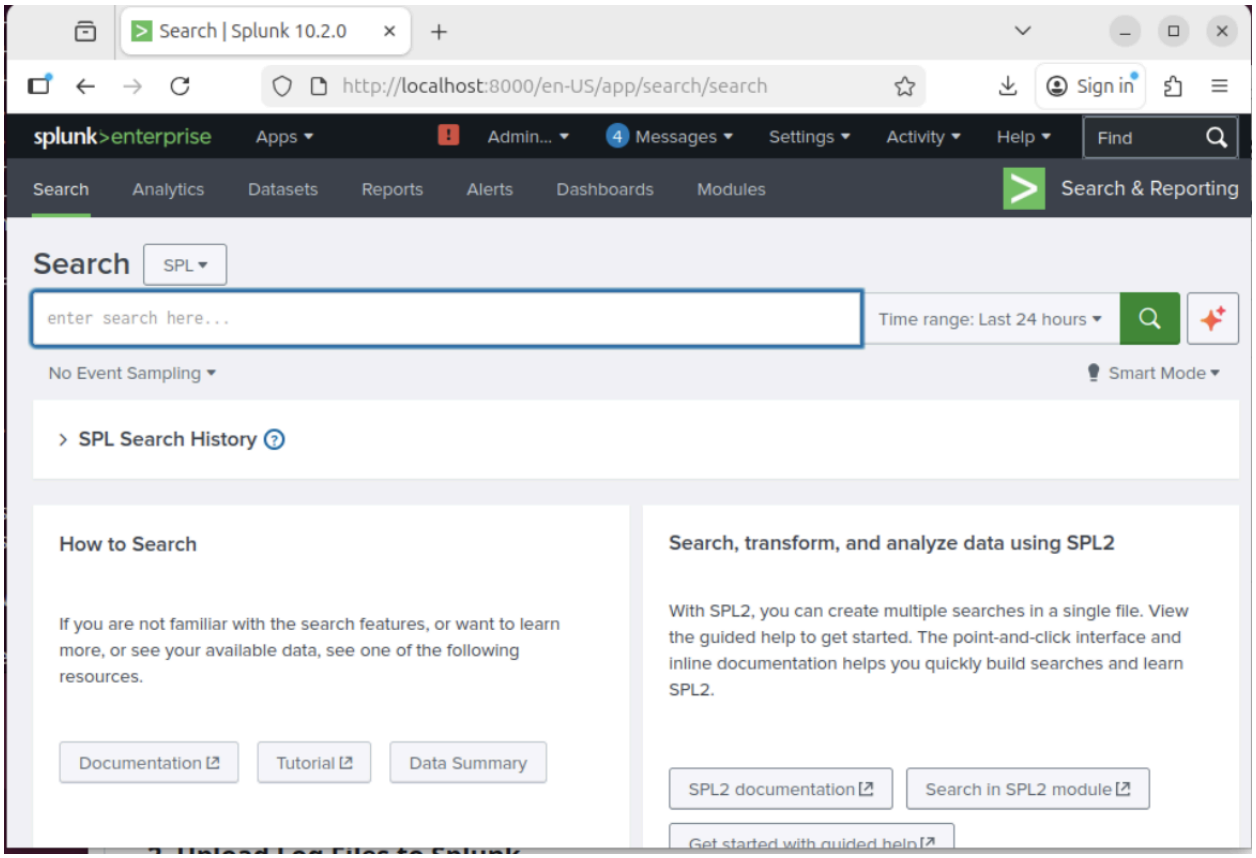
Threshold Value is Critical

The selection of the threshold value for failed login attempts was not an easy task. If the threshold value was set too low, there would be too many false positives, and if it was set too high, there would be a possibility of missing early attack indicators.



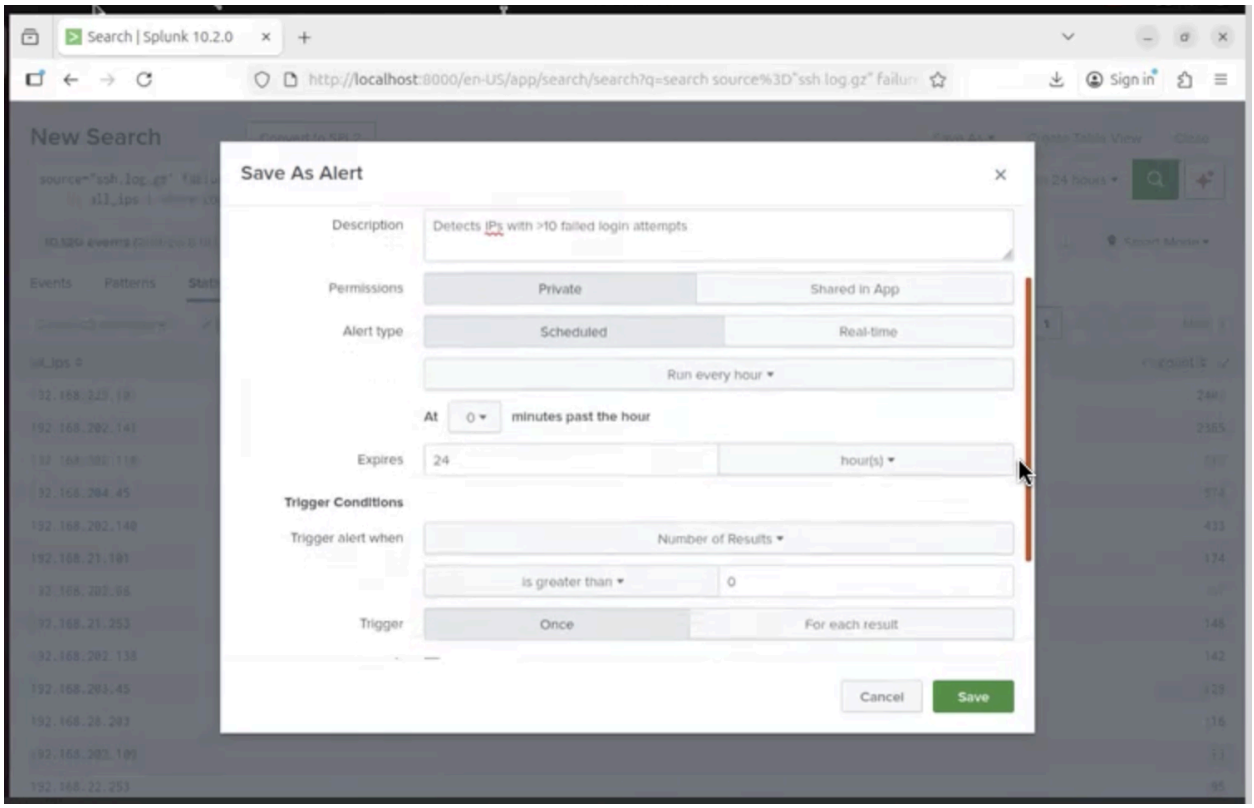
Visualization Adds to Situational Awareness

The development of the dashboard was very effective in situational awareness of attack patterns. Visualization of the top offending IPs and attack patterns over time helped in quick identification of malicious attacks and in the communication of results to both technical and non-technical audiences.



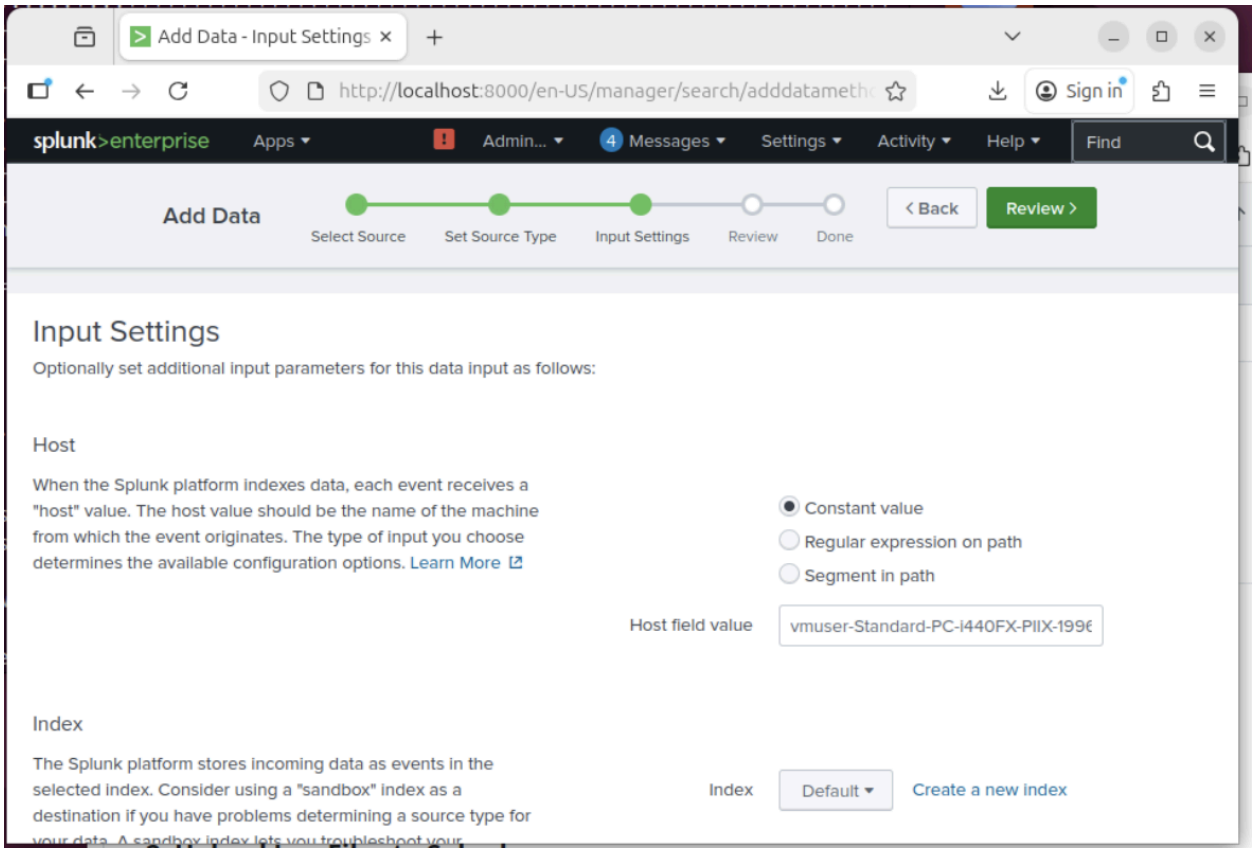
Automated Alerts: Proactive Defense

The development of automated alerts showed the effectiveness of proactive defense. Automated alerts enabled administrators to quickly react to malicious activity, thereby eliminating the need for manual log analysis. However, alert fatigue needs to be addressed by proper tuning and filtering.



Data Volume and Noise

Dealing with large amounts of log data emphasized the importance of efficient indexing and optimization of queries. Background noise from benign failed logins (such as password mistakes) also required additional filtering to ensure that the detection logic stayed on track and focused on malicious activity.



Future Enhancements

In future implementations, the following enhancements could be made to improve detection and analysis:

- Adding geolocation and reputation information for source IPs to focus on more dangerous attackers.
- Adding anomaly-based detection in addition to threshold-based detection to find more complex attack patterns.
- Adding monitoring for other authentication protocols in addition to SSH to provide a more complete security picture.

Conclusion

Overall, this project has reaffirmed the importance of accurate log parsing, well-crafted detection logic, informative visualizations, and timely alerts in successful cybersecurity monitoring. It has also illustrated the iterative nature of security analysis and the importance of ongoing improvement in both process and tooling.